ECGR 3120 — One-Page Design Recommendation

Magnetically Steered Cardiac Ablation — Dashia Boyarskiy — Dr. Joy

Conditional recommendation: 80 mT is the lowest of 40, 80 and 160 mT fields meeting a *chosen illustrative* minimum of 290 mm^3 electrical power-density threshold-volume proxy over the five tested commanded angles from 15° to 75°. This numerical target is neither handout-specified nor clinically validated.

Field	75° proxy, 12-cell	6/12/24-cell range	Pair power	Mount force
40 mT	257.0 mm^3	241.8–264.3 mm^3	∼0.73 kW	∼174 N
80 mT	310.5 mm^3	301.0–315.3 mm^3	∼2.93 kW	∼697 N
160 mT	343.1 mm^3	324.2–353.2 mm^3	∼11.7 kW	∼2.79 kN

Coil requirements for 80 mT: about 201 A per coil in the starter-kit 155-turn geometry (∼ 31,155 ampere-turns/coil), dissipation ∼1.46 kW per coil and mechanical attraction about 697 N.

Critical thermal drawback: the free-convection model predicts around **263 K temperature rise** at 80 mT. Uncooled operation is not acceptable under this model; cooling and structural supports require redesign and independent verification.

Physical interpretation: stronger field reduces commanded-versus-achieved angular error, enabling deeper seating and higher predicted electrical power-density proxy. However, losses and attraction rise quadratically with current at fixed geometry. The 160 mT option costs about four times the power and force of 80 mT for a comparatively modest additional proxy volume.

Limitations: finite-grid ranges are observed numerical sensitivity, not statistical confidence intervals; the seating-depth law is a supplied geometric assumption, not a patient-validated contact model. The model solves quasi-static conduction and a power-density cutoff, not tissue heating, perfusion, thermal dose or a clinically validated lesion. The 290 mm^3 threshold is a screening choice, not a medical requirement.

Validation: 29/29 supplied acceptance tests passed. The final design decision remains conditional on validating the performance threshold, cooling solution, mechanical mounting and contact/tissue assumptions.